

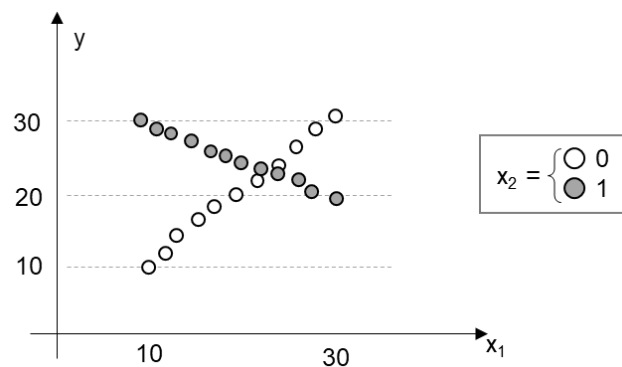
Exercise 1: Linear Regression I

Consider the following three points in an (x, y) plane: $(2, 2)$, $(3, 4)$, and $(8, 5)$.

- a) Using a solver (e.g. in MS Excel), compute the best regression line $y = a + bx$. Based on the actual and predicted values of this regression, what is the TSS, the ESS, and – based on these – the R^2 value?

Exercise 2: Linear Regression II

The following figure shows a schematic scatter plot of data points, the y -values of which are to be explained by different linear regression models based on x_1 and x_2 .



- a) In the following table, please indicate the coefficients for the respective regression model estimates (grey fields do not exist in the respective model).

	Model I	Model II	Model III	Model IV	Model V
<i>Intercept</i>					
x_1					
x_2					
$x_1 * x_2$					

Exercise 3: Spline Regression

Consider the dataset *splines.csv*, which shows annual income as well as the age of participants measured during a study.

- Analyze the relationship between income and age using linear, quadratic, as well as spline regression models.
- Assuming retirement affects the annual income of participants, estimate the retirement age based on the dataset.

Exercise 4: Logarithmic transformation

Consider the following small dataset of apartment prices in Berlin (prices in thousands of EUR, sizes in square meters). We will explore how a log transformation affects the shape and skewness of data distribution, outliers, variance and the interpretation of linear regression.

Apartment #	1	2	3	4	5	6	7	8	9	10	11	12
size (sqm)	40	55	60	70	80	90	100	120	140	160	180	220
price (kEUR)	276.7	249.8	309.1	479.8	488.1	264.7	469.2	417.9	485.3	622.8	649	1659

- Compute the **mean**, **median** and **skewness** for price and the natural logarithmic transformation of price. Visualize the values and interpret the results: How does the transformation affect the shape and spread of the distribution?
- Fit two simple linear regression models and calculate the estimated coefficients and R^2 for both:

Model A: $\text{price} = \alpha_A + \beta_A * \text{size}$

Model B: $\ln \text{price} = \alpha_B + \beta_B * \text{size}$

Briefly compare: Which model explains more variance? Which seems less influenced by the extreme price?

- Using **Model B**, interpret β_B in % terms: "Holding everything else constant, each additional 1 sqm increases price by about ____%". What are the approximate % changes in price for an additional 10 sqm and 20 sqm?
- When would you use a logarithmic transformation of price instead of price directly? When is that not appropriate?